Quantum State Discrimination: Detailed Analysis

Connection Between Cloning and State Estimation

1 Problem Statement

Quantum State Discrimination Problem

Topic: Quantum state discrimination and its connection to quantum cloning.
Statement: The task of learning or estimating an unknown quantum state is connected to cloning. The optimal strategy for estimating a state sometimes involves a "measure-and-prepare" scheme whose fidelity is bounded by the optimal cloning fidelity.
Questions to Address:

1. What is quantum state discrimination and why is it fundamentally limited?
2. How is quantum state discrimination related to quantum cloning?
3. What is a "measure-and-prepare" scheme?
4. What are the fidelity bounds for state discrimination?
5. How does the no-cloning theorem constrain state estimation?
6. What are the optimal strategies for discriminating quantum states?

2 Fundamental Concepts

2.1 Quantum State Discrimination

Key Concept: Quantum State Discrimination

Quantum State Discrimination is the task of determining which quantum state from a known set $\{|\psi_1\rangle, |\psi_2\rangle, \ldots, |\psi_n\rangle\}$ has been prepared, given a single copy (or multiple copies) of that state.
This is fundamentally different from classical state discrimination because:

- Quantum measurements are destructive
- Non-orthogonal states cannot be perfectly distinguished
- The measurement process disturbs the state

2.2 The No-Cloning Theorem

Fundamental Limitation: No-Cloning Theorem

The **No-Cloning Theorem** states that it is impossible to create an identical copy of an arbitrary unknown quantum state.
Mathematical Statement: There does not exist a unitary operator U such that:

$$U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle \tag{1}$$

for all quantum states $|\psi\rangle$.
Proof Sketch: Suppose such a U exists. For two non-orthogonal states $|\psi\rangle$ and $|\phi\rangle$:

$$U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle \tag{2}$$
$$U(|\phi\rangle \otimes |0\rangle) = |\phi\rangle \otimes |\phi\rangle \tag{3}$$

Taking the inner product and using unitarity:

$$\langle\psi|\phi\rangle = [\langle\psi|\phi\rangle]^2 \tag{4}$$

This implies $\langle\psi|\phi\rangle = 0$ or 1, meaning states must be orthogonal or identical. Contradiction!

2.3 Quantum State Fidelity

The **fidelity** between two quantum states ρ and σ is defined as:

$$F(\rho, \sigma) = \left[\mathrm{Tr}\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}\right]^2 \tag{5}$$

For pure states $|\psi\rangle$ and $|\phi\rangle$:

$$F(|\psi\rangle, |\phi\rangle) = |\langle\psi|\phi\rangle|^2 \tag{6}$$

The fidelity measures how "close" two quantum states are, with $F = 1$ for identical states and $F = 0$ for orthogonal states.

3 Connection Between Cloning and State Discrimination

Key Insight: Cloning-Discrimination Connection

The connection between quantum cloning and state discrimination arises because:

1. **Perfect discrimination requires perfect cloning:** If we could perfectly clone an unknown state, we could perform measurements on multiple copies to determine the state with arbitrary precision.
2. **Optimal discrimination is limited by optimal cloning fidelity:** Since perfect cloning is impossible, the best we can do is approximate cloning, which limits our ability to discriminate states.
3. **Measure-and-prepare schemes:** These schemes effectively create "imperfect clones" of the original state, with fidelity bounded by fundamental quantum limits.

3.1 Optimal Cloning Fidelity

For a universal quantum cloning machine that produces M copies from N input copies, the optimal fidelity is:

$$F_{\text{clone}}(N \to M) = \frac{N(d+1) + M - N}{M(d+1)} \tag{7}$$

where d is the dimension of the Hilbert space.

For qubits ($d = 2$), cloning $1 \to 2$ (one copy to two copies):

$$F_{\text{clone}}(1 \to 2) = \frac{1(2+1) + 2 - 1}{2(2+1)} = \frac{4}{6} = \frac{2}{3} \approx 0.833 \tag{8}$$

Important Result

The $5/6$ bound: For optimal cloning of a single qubit into two copies, each copy has fidelity $F = 5/6$ with the original state.

4 Measure-and-Prepare Schemes

4.1 Description of the Scheme

Measure-and-Prepare Strategy

A **measure-and-prepare** scheme for state discrimination works as follows:

1. **Measure:** Perform a measurement on the unknown state $|\psi\rangle$, obtaining outcome m with some probability
2. **Prepare:** Based on the measurement outcome m, prepare a new state $|\phi_m\rangle$
3. **Estimate:** Use the prepared state as an estimate of the original state

This is essentially an imperfect cloning procedure, where we try to recreate the original state based on partial information from the measurement.

4.2 Mathematical Description

Let the unknown state be $|\psi\rangle$. We perform a POVM (Positive Operator-Valued Measure) with elements $\{E_m\}$ satisfying:

$$\sum_m E_m = \mathbb{I} \tag{9}$$

The probability of outcome m is:

$$p(m|\psi) = \langle\psi| E_m |\psi\rangle \tag{10}$$

Based on outcome m, we prepare state $|\phi_m\rangle$. The average fidelity is:

$$\bar{F} = \sum_m p(m|\psi) |\langle\psi|\phi_m\rangle|^2 \tag{11}$$

5 Fidelity Bounds for State Discrimination

5.1 General Bounds

For discriminating between n pure states $\{|\psi_i\rangle\}$ with prior probabilities $\{p_i\}$:

Fidelity Bounds

Lower Bound (Random Guessing):

$$F_{\min} = \sum_i p_i^2 \tag{12}$$

Upper Bound (Related to Cloning): For measure-and-prepare schemes with single-copy measurements, the fidelity is bounded by:

$$F_{\text{measure-prepare}} \leq F_{\text{optimal cloning}} \tag{13}$$

For qubits with uniform prior:

$$F \leq \frac{2}{3} \quad \text{(optimal cloning bound)} \tag{14}$$

5.2 Specific Case: Two Pure States

Consider discriminating between two non-orthogonal pure states $|\psi_1\rangle$ and $|\psi_2\rangle$ with inner product:

$$\langle\psi_1|\psi_2\rangle = \cos\theta \tag{15}$$

Helstrom Bound: The maximum probability of correct discrimination is:

$$P_{\text{correct}}^{\max} = \frac{1}{2}\left(1 + \sqrt{1 - |\langle\psi_1|\psi_2\rangle|^2}\right) = \frac{1}{2}(1 + \sin\theta) \tag{16}$$

Minimum Error Probability:

$$P_{\text{error}}^{\min} = \frac{1}{2}(1 - \sin\theta) \tag{17}$$

Interpretation

- If $|\psi_1\rangle \perp |\psi_2\rangle$ (orthogonal), then $P_{\text{error}} = 0$ (perfect discrimination)
- If $|\psi_1\rangle = |\psi_2\rangle$ (identical), then $P_{\text{error}} = 1/2$ (no information)
- For non-orthogonal states, there is always some error probability

6 Optimal Strategies for State Discrimination

6.1 Minimum Error Discrimination (Helstrom Measurement)

For minimizing the probability of error when discriminating between states $\{|\psi_i\rangle\}$ with priors $\{p_i\}$:

Optimal Measurement: The optimal POVM is obtained from the spectral decomposition of:

$$\Gamma = \sum_i p_i |\psi_i\rangle \langle\psi_i| \tag{18}$$

The measurement operators are projectors onto eigenspaces where:

$$\Pi_i = \text{Projector onto eigenvectors with eigenvalue} > \lambda_{\text{threshold}} \tag{19}$$

6.2 Unambiguous State Discrimination (USD)

Alternative Strategy: Unambiguous Discrimination

In **Unambiguous State Discrimination**, we allow for inconclusive results but require that when we do get a conclusive result, it is always correct.
Measurement Outcomes:

- "State 1" (with certainty)
- "State 2" (with certainty)
- "Don't know" (inconclusive)

For two states with $\langle\psi_1|\psi_2\rangle = \cos\theta$:

$$P_{\text{inconclusive}}^{\text{min}} = \cos\theta \tag{20}$$

$$P_{\text{success}} = 1 - \cos\theta = 2\sin^2(\theta/2) \tag{21}$$

6.3 Maximum Confidence Discrimination

Maximize the probability of correct identification when a conclusive answer is obtained:

$$P(\text{correct}|\text{conclusive}) = \frac{P_{\text{correct}}}{P_{\text{conclusive}}} \tag{22}$$

7 Detailed Example: Discriminating Two Qubit States

7.1 Problem Setup

Consider two pure qubit states:

$$|\psi_1\rangle = |0\rangle \tag{23}$$
$$|\psi_2\rangle = \cos\theta\,|0\rangle + \sin\theta\,|1\rangle \tag{24}$$

with equal prior probabilities $p_1 = p_2 = 1/2$ and $0 < \theta < \pi/2$.

7.2 Measure-and-Prepare Scheme

Step 1 - Measurement: Perform a measurement in some basis. Let's try the computational basis $\{|0\rangle, |1\rangle\}$.

Measurement outcomes:

- If we measure $|0\rangle$: Could be $|\psi_1\rangle$ (probability 1) or $|\psi_2\rangle$ (probability $\cos^2\theta$)
- If we measure $|1\rangle$: Must be $|\psi_2\rangle$ (probability $\sin^2\theta$)

Step 2 - Prepare: Based on outcome, prepare:

$$\text{Outcome } |0\rangle \to \text{Prepare } |\phi_0\rangle = \alpha|0\rangle + \beta|1\rangle \tag{25}$$
$$\text{Outcome } |1\rangle \to \text{Prepare } |\phi_1\rangle = |\psi_2\rangle \tag{26}$$

Step 3 - Calculate Fidelity:
Average fidelity:

$$\bar{F} = \frac{1}{2}[\text{Fidelity from } |\psi_1\rangle] + \frac{1}{2}[\text{Fidelity from } |\psi_2\rangle] \tag{27}$$

For $|\psi_1\rangle$:

$$F_1 = |\langle 0|\phi_0\rangle|^2 = |\alpha|^2 \tag{28}$$

For $|\psi_2\rangle$:

$$\begin{aligned} F_2 &= \cos^2\theta \cdot |\langle\psi_2|\phi_0\rangle|^2 + \sin^2\theta \cdot |\langle\psi_2|\phi_1\rangle|^2 && (29) \\ &= \cos^2\theta \cdot |\alpha\cos\theta + \beta\sin\theta|^2 + \sin^2\theta \cdot 1 && (30) \end{aligned}$$

The optimal choice of $|\phi_0\rangle$ maximizes the average fidelity subject to the constraint that it must be a valid quantum state.

7.3 Optimal Fidelity

For this specific case with optimal measurement and preparation strategy:

$$\boxed{F_{\text{optimal}} = \frac{1+\cos\theta}{2}} \tag{31}$$

This can be compared to:

- Random guessing: $F_{\text{random}} = \frac{1+\cos^2\theta}{2}$
- Cloning bound: $F_{\text{clone}} = \frac{2}{3}$ (for universal cloning)

8 Connection to Quantum Information Theory

Information-Theoretic Perspective

The connection between cloning and discrimination can be understood through information theory:

1. Information Content:

- A single qubit contains infinite classical information (continuous parameters θ, ϕ on Bloch sphere)
- But we can only extract finite information through measurement

2. Information-Disturbance Tradeoff:

$$\text{Information Gained} + \text{State Disturbance} \geq \text{Constant} \tag{32}$$

3. Holevo Bound: The maximum classical information extractable from n qubits is:

$$\chi = S(\rho) - \sum_i p_i S(\rho_i) \leq n \text{ bits} \tag{33}$$

where $S(\rho) = -\text{Tr}(\rho \log \rho)$ is the von Neumann entropy.

9 Applications and Implications

9.1 Quantum Cryptography

Security from No-Cloning:

- An eavesdropper cannot clone quantum states to extract information without detection
- State discrimination limits what information can be extracted from intercepted qubits
- BB84 protocol relies on the impossibility of perfectly discriminating non-orthogonal states

9.2 Quantum State Tomography

Full State Reconstruction:

- Requires multiple copies of the state
- Number of measurements scales with dimension: $O(d^2)$ for dimension d
- Each measurement provides limited information due to discrimination bounds

9.3 Quantum Machine Learning

Pattern Recognition:

- Training data encoded in quantum states
- Classification requires discriminating between quantum states
- Fidelity bounds limit classification accuracy

10 Summary of Key Results

Summary: Cloning-Discrimination Connection

1. Fundamental Limitations:

- No-cloning theorem prevents perfect state copying
- Non-orthogonal states cannot be perfectly discriminated
- These limitations are two sides of the same coin

2. Fidelity Bounds:

- Optimal cloning fidelity (qubits): $F = 5/6$ per clone for $1 \to 2$
- Measure-and-prepare fidelity: $F \leq F_{\text{clone}}$
- Universal bound relates cloning and discrimination performance

3. Optimal Strategies:

- Minimum error discrimination (Helstrom bound)
- Unambiguous discrimination (allows inconclusive results)
- Maximum confidence discrimination
- Choice depends on application requirements

4. Key Equation: For measure-and-prepare schemes discriminating/estimating quantum states:

$$\boxed{F_{\text{discrimination}} \leq F_{\text{optimal cloning}} \leq 1} \tag{34}$$

The equality $F = 1$ holds only for orthogonal states or when we have additional copies of the state.

11 Advanced Topics

11.1 Probabilistic Cloning

While deterministic perfect cloning is impossible, **probabilistic cloning** with some success probability $p < 1$ can achieve perfect fidelity when successful:

$$U(|\psi\rangle \otimes |0\rangle) = p\,|\psi\rangle \otimes |\psi\rangle + \sqrt{1-p^2}\,|\Phi_\perp\rangle \tag{35}$$

where $|\Phi_\perp\rangle$ is orthogonal to $|\psi\rangle \otimes |\psi\rangle$.

11.2 Asymmetric Cloning

In asymmetric cloning, different copies can have different fidelities:

$$F_1 \neq F_2, \quad \text{but} \quad F_1 + F_2 = \text{constant} \tag{36}$$

This provides flexibility in applications where one copy needs higher fidelity than another.

11.3 State-Dependent Cloning

For a known set of states rather than universal cloning, better fidelities can be achieved:

$$F_{\text{state-dependent}} > F_{\text{universal}} \tag{37}$$

12 Conclusion

The relationship between quantum cloning and state discrimination reveals deep connections in quantum information theory:

1. Both are fundamentally limited by the same quantum mechanical principles
2. The no-cloning theorem directly implies limits on state discrimination
3. Measure-and-prepare schemes provide a practical link between these concepts
4. Optimal strategies must balance information extraction with state preservation
5. These limits enable quantum cryptography while constraining quantum computing

The fidelity bounds derived from optimal cloning serve as fundamental benchmarks for any state estimation or discrimination protocol, highlighting the intrinsic quantum nature of information.